

Data Structures: Trees and Graphs

Subject: Computer Science

Trees

A tree is a hierarchical, non-linear data structure with a root node and child nodes. Each node has exactly one parent (except the root). Trees are acyclic connected graphs.

- Binary Tree: Each node has at most 2 children.
- Binary Search Tree (BST): For every node, left subtree values < node value < right subtree values.
- Balanced BST: Height is $O(\log n)$, ensuring efficient search, insert, and delete.

Graphs

A graph $G = (V, E)$ consists of vertices V and edges E connecting them. Graphs can be:

- Directed (digraph) or Undirected
- Weighted or Unweighted
- Cyclic or Acyclic
- Connected or Disconnected

Trees are a special case of acyclic connected undirected graphs.

Key Algorithms

BFS (Breadth-First Search):

- Uses a queue; explores level by level.
- Time: $O(V + E)$ Space: $O(V)$
- Use for: shortest path in unweighted graphs.

DFS (Depth-First Search):

- Uses a stack (or recursion); explores as far as possible before backtracking.
- Time: $O(V + E)$ Space: $O(V)$
- Use for: cycle detection, topological sort.

Dijkstra's Algorithm:

- Finds shortest path in weighted graphs (non-negative weights).
- Time: $O((V + E) \log V)$ with a min-heap.

Exam Topics

1. Perform in-order, pre-order, and post-order traversals on a given binary tree.
2. Compare BFS and DFS and state when you would use each.
3. Trace Dijkstra's algorithm on a small weighted graph.
4. Detect a cycle in directed and undirected graphs.
5. What is the time complexity of inserting into a balanced vs. unbalanced BST?